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Study and Field Trial of Discontinuous Chemical Flooding for Mature Offshore Oilfield with Ultra High Water Cut

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Abstract

Large portion of offshore oilfield in China are mature oilfields with ultra high water cut and recovery rate. The fragmentary distributed remaining oil and aged downhole equipment bring great challenges to the sustainable oil production. A new chemical flooding technique called discontinuous chemical flooding (DCF) was developed to achieve effective remaining oil extraction and has been implemented in oilfield Q, a typical mature oilfield with a water cut of 95% and a recovery rate of more than 40%.

Different chemical agent slugs were designed to deal with channeling effects and injectivity problem. Two great challenges were encountered during the field implementation. The injection pressure rose sharply by 6.6 MPa within two months after polymer injection. Acidizing, re-perforation, and alternating injection of polymer with different molecular weights (MW) were implemented for injectivity improvement. Hydrogen sulfide (H_2S) was detected from the platform, and the polymer viscosity was greatly affected. Bactericide was added to the production system to reduce the concentration of H_2S. Nitrogen replacement method was used in the polymer dispensing equipment to block the contact between polymer and oxygen.

A total of 13 acidizing treatments were performed since the project start. The apparent injectivity index improvements with a maximum value of 300% were achieved. The re-perforation treatment was conducted in Well I6. Deep penetration perforation was conducted from the original well section with a density of 20 SPM and a longitudinal extension of 6.1 meters. Significant improvements were observed in both injectivity and pressure rise rate. The water injection rate was increased from 100 m3/d to 445 m3/d at polymer concentration of 800 mg/L, and the stable injection was maintained for 8 months. A moderate MW polymer was injected for trial. The average pressure rise rate was reduced by 0.097MPa/d. Under non-plugging conditions, the polymer with lower MW yields decent injectivity improvements. The wellhead polymer

viscosity was reduced by 42% due to H_2S. Numerical simulation inversion indicates that the in-situ polymer viscosity is only 2.8 mPas. After H_2S treatment, the concentration in the production system was reduced from 50ppm to less than 15ppm. The wellhead polymer viscosity was recovered.

Within 2 years of agent adjustment, well intervention and H_2S treatment, 11 out of 17 wells had great oil incremental performance. The incremental oil production is evaluated to be 1.69×10^4 m^3. The field trail proves the technical feasibility of enhancing oil recovery through DCF in mature offshore oilfields with ultra high water cuts.

Introduction

The Bohai Oilfield, with an annual crude oil production approaching 40 million tons, ranks among China's largest offshore crude oil production bases. Compared with onshore oilfields, the offshore oilfields predominantly consist of unconsolidated sandstone reservoir, characterize as high porosity and high permeability with great formation heterogeneity and high crude oil viscosity. They are mainly developed by unfavorable well patterns, large well spacing, large segments of multi-layer combined injection and production, as well as high rate of injection and production (Zhang et al., 2023). Besides, most of them have entered the high water cut stage. With years of waterflooding development, the intensified reservoir heterogeneity and channeling effects have significantly increased the technical challenges for the recovery of remaining oil reserves.

The polymer-dominated chemical flooding has proven to be an effective enhanced oil recovery (EOR) technique for offshore oilfields (Li et al., 2017). This approach has become one of the most effective EOR techniques for medium-high permeability reservoirs in China. Laboratory experiments and field trials demonstrate that polymer flooding typically achieves about 10% incremental oil recovery compared with waterflooding. Nevertheless, the conformance improvement capability of polymers remains limited, facilitating bad performance when dealing with thief zone. Additionally, high injection pressures were always observed in continuous polymer injection during which the sand production risk get increased. These challenges necessitate the development of alternative EOR technologies to further improve offshore oilfield recovery factors.

In order to achieve high oil recovery for mature oilfield in high water cut stage, a new chemical flooding method called discontinuous chemical flooding (DCF) was proposed and has been applied in a field trial. In DCF, dynamic reservoir monitoring techniques such as injectivity index test, pressure fall-off test and tracer test are utilized to evaluate time dependent reservoir characteristics and flow patterns (Jian et al., 2023). Based on real-time data, the customized chemical agents with distinct function are selectively injected to the target zones. This approach effectively regulates abrupt mobility changes, mitigates formation heterogeneity, and enhances volumetric sweep efficiency while maintaining optimal injectivity, ultimately achieving precision injection-production optimization and ensuring sustained high-performance recovery throughout the entire production period (Gao et al., 2021, Yu et al., 2021). In general, the DCF employs a diverse type of chemical agents and they can be defined as four types of chemical slug, which are front slug, main slug, secondary slug, and auxiliary slug (Zhang et al., 2024). Before the injection of the main slug, the front slug consisting the adsorption reducers and strong conformance control agents are injected to enhance the injectivity and improve the injection profiles of the near-wellbore region. The secondary slug is designed to displace the remaining oil in relatively low-permeability zones that are inaccessible to the main slug. Moreover, if the injection rates fall below requirements, an auxiliary slug containing injectivity enhancer or water is introduced to restore the injectivity in the near-wellbore region.

To verify the technical feasibility oil incremental performance of DCF, a full field DCF scheme was designed and implemented in oilfield Q in Bohai Bay. It is a typical high porosity and high permeability reservoir with an average porosity of 30% and an average permeability of 1300 mD. With a low oil viscosity of around 5 cP, the field has been primarily developed through waterflooding over 20 years. The field has entered the ultra-high water-cut stage, with water cut reaching 95% and recovery rate exceeding 42%. The economic limit year is evaluated to be 2026. During the field operation, two great challenges were encountered. Abnormal rises in injection pressure were observed and H_2S was detected, which greatly affected the injection volume requirements and polymer performance, respectively. In this paper, the causing factors, treatment strategies and field trial performance with DCF are systematically investigated.

Scheme Overview

The proposed development scheme for Oilfield Q involves 12 injection wells and 17 production wells. The total injection volume is 0.409 PV. With DCF slug design, the project employs a multi-slug injection mode including 4 types of slugs which are front, main, secondary and auxiliary slugs. 10 wells of front slug were designed based on the thief zone analysis. The injection concentrations and slug sizes of the front slug and the main slug for different injectors were optimized using CMOST module in CMG. 23 wells of secondary slug treatments were designed based on the simulation prediction of the water cut and the produced chemical agent concentration. The auxiliary slug consists of low-concentration polymer, injection enhancement, and water. It was designed to be injected when the injection rate fails to meet the designed injection target. The DCF scheme is projected to have an EOR value of 5.57% and a maximum water cut reduction of 6.7%. Moreover, the project evaluation indicates that the economic limit year will be extend from the year of 2026 to the year of 2029. The schematic diagram of oil production rate for chemical flooding scheme is shown in Fig. 1.

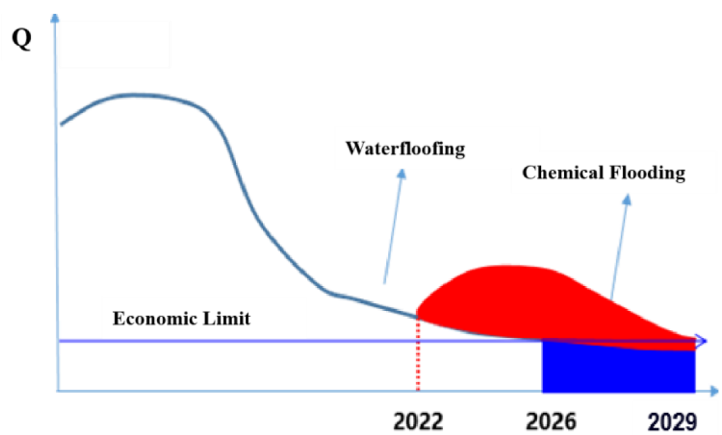

Figure 1—Schematic diagram of oil production rate for chemical flooding scheme.

The Implementation Process of the Field Trial

The chemical flooding implementation was carried out in four operational batches. Among the 12 designated injection wells, five were initiated on schedule, six were delayed by 2-5 months, and one experienced a 12-month delay in chemical agent injection. The completion status of the designed scheme can be found in Table 1. As of the current stage, excluding the impact of implementation delays, the project should have injected 0.138 PV with 3389 tons of polymer. However, the actual injection has achieved only 0.106 PV with 2094 tons of polymer utilized. The actual operations achieved only 76.8% of the target PV injection and 61.8% of the planned polymer consumption.

Table 1—Comparison between actual injection and scheme design.

	Scheme Design (based on the actual injection time)	Current actual Implementation	Completion Rate (based on the scheme of actual injection time)
Number of Injection well	12	12	100%
Pore Volume	0.138	0.106	76.8%
Polymer Mass	3389	2094	61.8%

Problems Encountered and Overcome Strategies

The rapidly risen injection pressure

Prior to polymer injection, the 12 injectors had an average injection pressure of 4.9 MPa. Within 1-2 months of polymer flooding, the average pressure increased by 6.6 MPa and reached 11.5 MPa. 7 wells reached the upper pressure limit of 12.5 MPa even when the injection concentrations are below the designed values. The rapid pressure buildup during early-stage polymer injection resulted in the insufficient injection for some wells. In this situation, the originally planned 33 wells of gel-based front slugs and secondary slugs proved difficult to implement in the actual field operation. Selecting the stage when each well reaches the maximum injection pressure under constant injection rate after polymer flooding, the theoretical injection pressure rise value is calculated using the equation bellow.

$$\Delta P = 0.0018\left[\frac{q}{\frac{kh}{\mu}}\ln\left(\frac{r}{r_w}\right)\right]$$

Where q is injection rate, m^3/d. k is formation permeability, μm2. h is the effective thickness, m. μ is polymer viscosity, mPa·s. r is the polymer front radius, m. r_w is the wellbore radius, m.

The theoretical and actual injection pressure rise values for all injection wells except I6, which is a well with great injectivity issue in waterflooding stage are plotted in Fig. 2. It implies that the actual pressure rise values are way higher than the theoretical values except that of injector I11. Since the average polymer propagation distance is calculated to be 27m, it is speculated that plugging was generated in the near wellbore region, which is a typical formation damage that impair well injectivity (Soroush et al., 2021).

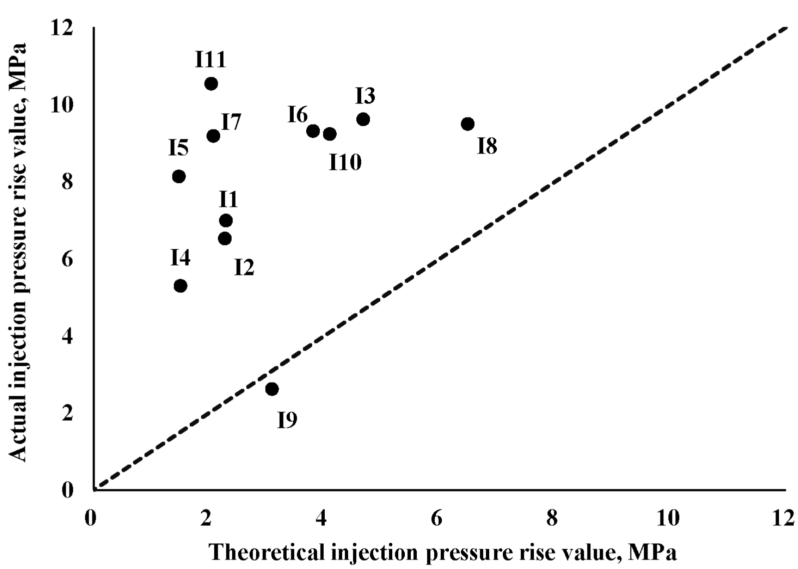

Figure 2—Comparison of actual and theoretical injection pressure rise values for 11 injectors.

To address the sharply risen injection pressure, a comprehensive measure was adopted. First, the filtration precision of the polymer stock solution filter mesh was increased from 120 mesh to 200 mesh to remove "fish eyes" generated during the preparation of the stock solution. Subsequently, acidizing, re-perforation, and alternating injection of polymer with different MWs were implemented for individual wells based on their specific downhole conditions.

Acidizing Treatment. To address this issue, acidizing measure was implemented. Since the project initiation, a total of 13 well acidizing treatments has been conducted. These treatments demonstrated an average effective duration of 224 days. Compared with the injectivity index measured 30 days prior to the well intervention, the acidizing treatments resulted in varying degrees of improvement in the apparent injectivity index among all treated wells, with the maximum enhancement of 300%. The improvement can be seen in Fig. 3.

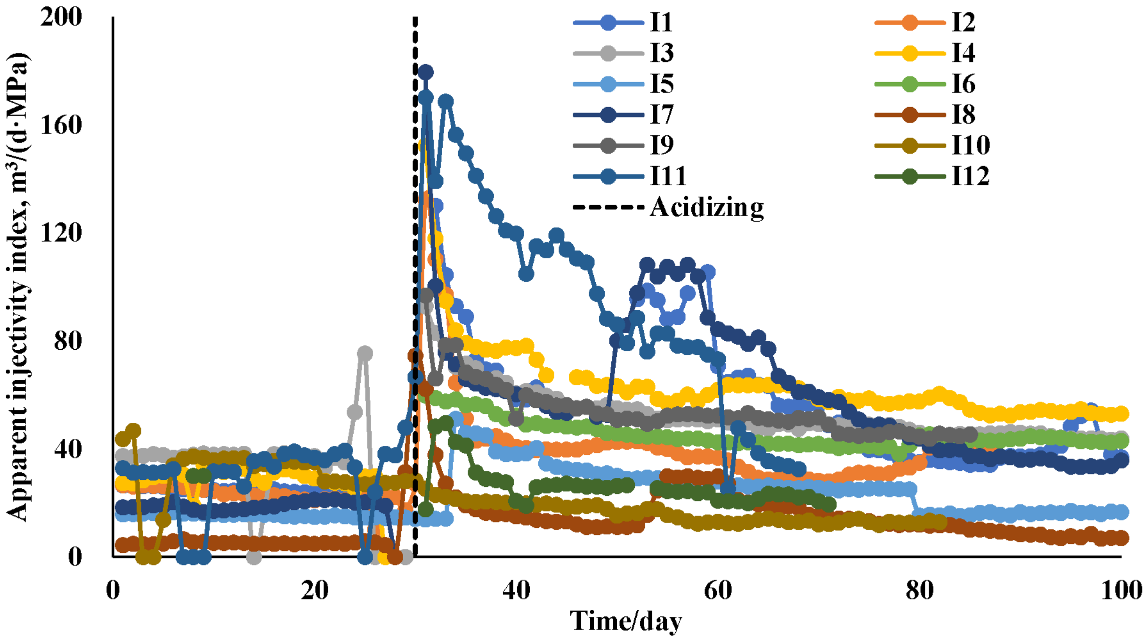

Figure 3—The apparent injectivity index after acidizing treatment for 12 injectors.

Re-perforation. During implementation, acidizing stimulation measures still failed to resolve the injection issues in certain wells. For example, Well I6, located in the central area of the oilfield, is surrounded by five producers. The well was completed with a wire-wrapped screen+gravel pack for sand control, targeting three perforated intervals with a total net pay thickness of 14.7 m. It was converted to an injector one year before the chemical flooding. The injection rate and pressure can be seen in Fig. 4. It initially achieved an injection rate of 480 m^3/day, but rapidly declined to 240 m^3/day. Three subsequent acidizing treatments temporarily restored peak injection to 680 m^3/day, but the rate quickly dropped again. Due to the injectivity issue, a 300 m^3/day injection shortfall was observed during polymer injection.

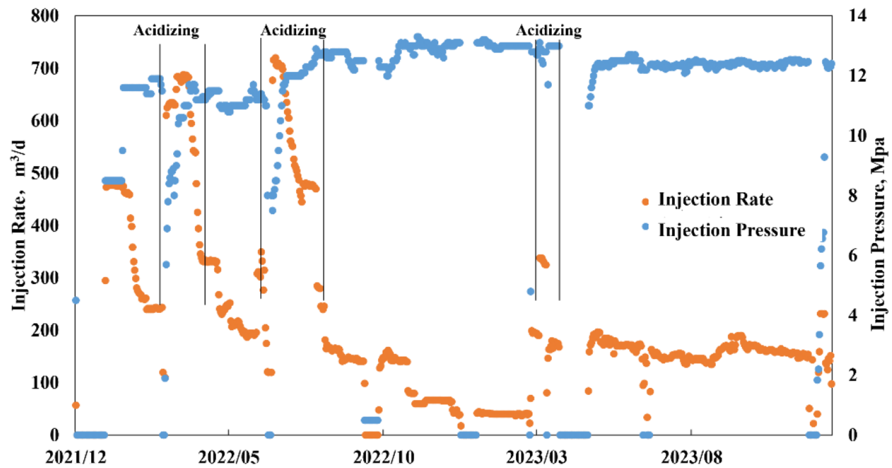

Figure 4—The injection rate and pressure for well I6 with multiple acidizing treatments.

To address this issue, a reperforation operation was conducted on Well I6, employing a shot density of 20 shots per meter (SPM) within the original perforation intervals. The operation achieved an enhanced penetration depth of 852mm while extending the perforated zone vertically by 6.1m. After reperforation, the well was put into injection without sand control measures, adopting a commingled injection approach. The intervention demonstrated significant improvements in both injectivity and pressure buildup rate. Subsequently, as shown in Fig. 5, the well maintained a stable injection for 8 months at a polymer concentration of 800 mg/L, consistently meeting the target injection rate of 445 m^3/day.

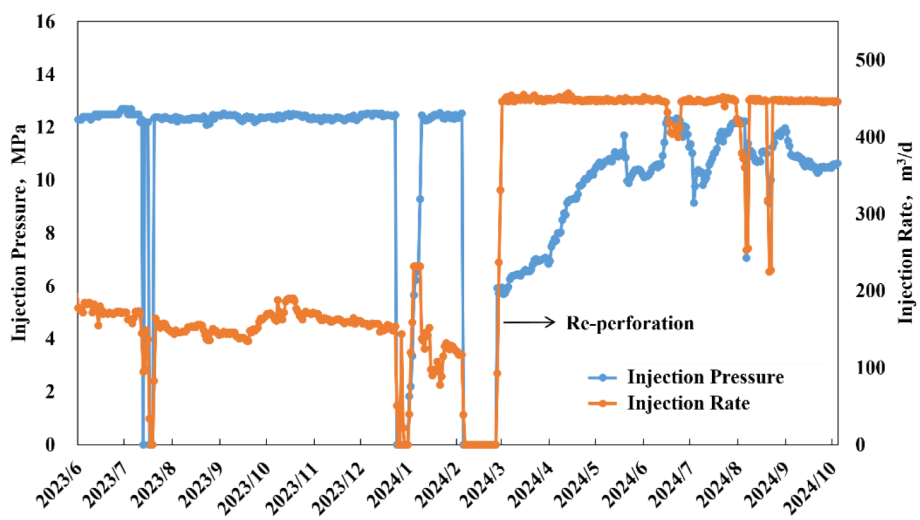

Figure 5—The injection rate and pressure for well I6 after re-perforation.

Alternating Injection of Polymer with Different MW. While the acidizing and reperforation treatments are only effective for removing formation damage in the near-wellbore and wellbore region, the in-depth polymer mobility is fundamentally governed by the polymer-reservoir compatibility. The polymer MW is a key factor that affecting the polymer-reservoir compatibility. In DCF scheme, different agents get involved to deal with problems encountered during chemical flooding. To deal with injectivity issue, polymers with moderate MWs were carried out for lab evaluation and field trial.

Polymer solutions with varying MWs (8 million, 12 million, 14 million,16 million, 18 million, and 21 million) were selected for coreflooding experiments. The result can be seen in Table 2. demonstrates that at a constant concentration of 1200 mg/L, the polymer with 16-million MW exhibits a lower injection pressure compared to the polymer with 21-million MW, though with a 3.3% reduction in oil displacement efficiency.

When maintaining an injected viscosity of 30 mPa·s, the polymers with different MWs achieves similar EOR values, all exceeding 10%, at equivalent pore volumes injected. However, polymers with lower MV required higher concentrations to reach the target viscosity of 30 mPa·s.

Table 2—Coreflooding results for polymer with different MWs.

Agent	Concentration (mg/L)	Viscosity (MPa·s)	Injection Pressure (MPa)	Recovery rate (waterflooding)	Recovery rate (polymer flooding)	Recovery rate (followed up waterflooding)	EOR (%)
Water	/	/	/	/	/	45.84	/
P01-2100	1200	30.9	0.358	40.24	53.31	57.74	11.90
P01-1800	1200	25.2	0.203	42.13	52.79	56.72	10.88
P01-1600	1200	19.6	0.242	41.30	51.04	54.43	8.61
	1500	30.3	0.310	41.21	53.88	57.71	11.87
P01-1400	1200	14.5	0.18	40.16	51.56	53.36	7.52
	1700	30.8	0.284	39.92	52.29	57.25	11.41
P01-1200	1200	12.2	0.125	40.78	48.88	52.11	6.27
	2000	30.8	0.201	40.26	51.58	56.95	11.09
P01-800	1200	7.8	0.100	39.51	46.72	51.56	5.72
	2500	30.1	0.186	38.69	52.13	55.29	9.45

Key properties measurements were conducted on the polymer with 16-million MW and 21-million MW. The result can be seen in Fig. 6 and Table 3. The lab test indicates that at a concentration of 1200 mg/L, the polymer with 16-million MW exhibits a 30% reduction in viscosity, a 60% decrease in residual resistance factor, a 100% improvement in injectivity, and a 10% increase in post-shear viscosity retention when compared to the polymer with 21-million MW.

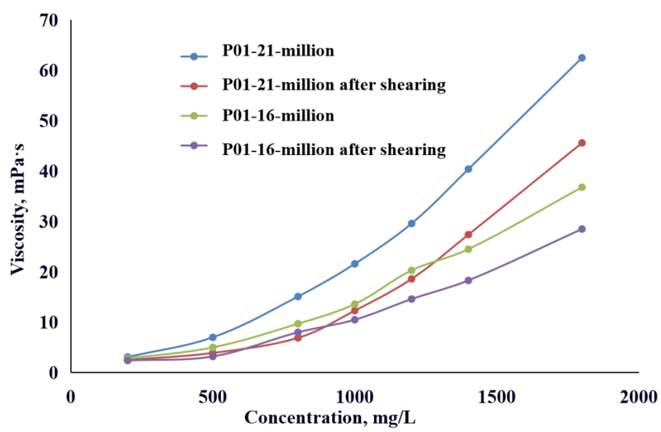

Figure 6—Viscosity of polymers with 16-million MW and 21-million MW.

Table 3—Properties measurement for polymer with different MWs.

Properties	Polymer with 21-million MW	Polymer with 16-million MW
Apparent viscosity at concentration of 1200 m/L (mPa·s)	34.9	20.5
Viscosity retention rate after shearing (%)	77	83
Residual resistant factor	5.1	2.0

Given that the polymer with 21-million MW demonstrates superior oil displacement capability compared to the polymer with 16-million MW, an alternating injection strategy utilizing polymers with different MWs was proposed. The polymer with 16-million MW was put into a field trial for appropriately two months. The injection pressure curves are shown in Fig. 7. The injection of the polymer with 21-million MW exhibited a pressure increase rate of 0.328 MPa/d, while the polymer with 16-million MW showed a relatively lower rate of 0.231 MPa/d, representing an average reduction of approximately 0.097 MPa/day. Subsequently, all injectors were subjected to alternating injection of 16-million and 21-million MW polymers, with water injection intermittently inserted between polymer slugs. The injection pressure curves indicate a notable decline in the average injection pressure during this alternating injection period.

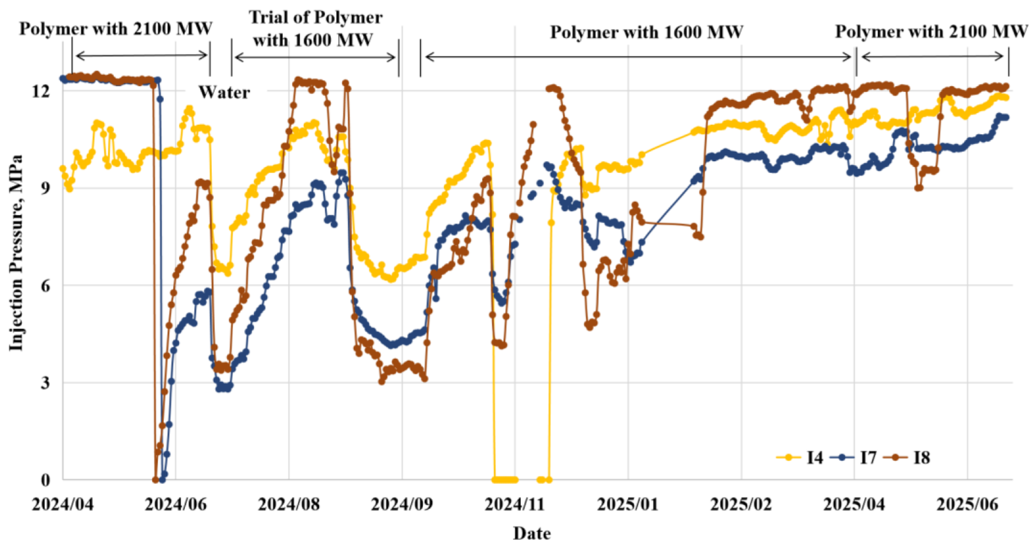

Figure 7—The injection pressures of 3 typical injector with different MW polymers injection.

The injection trial results from 12 injectors revealed two distinct behavioral patterns. The apparent injectivity index before and after 16-million MW polymer injection can be seen in Fig. 8. For wells achieving target injection rates, polymers with 16 million MW demonstrated decent injectivity improvements, with the enhancement effect diminishing at higher injection pressures. In contrast, wells failing to reach target injection rates, for example I3 and I12 showed no injectivity improvement regardless of polymer selection. These findings demonstrate that under non-plugging conditions, the polymer with 16-million MV can effectively enhance the well injectivity and delay the pressure build up.

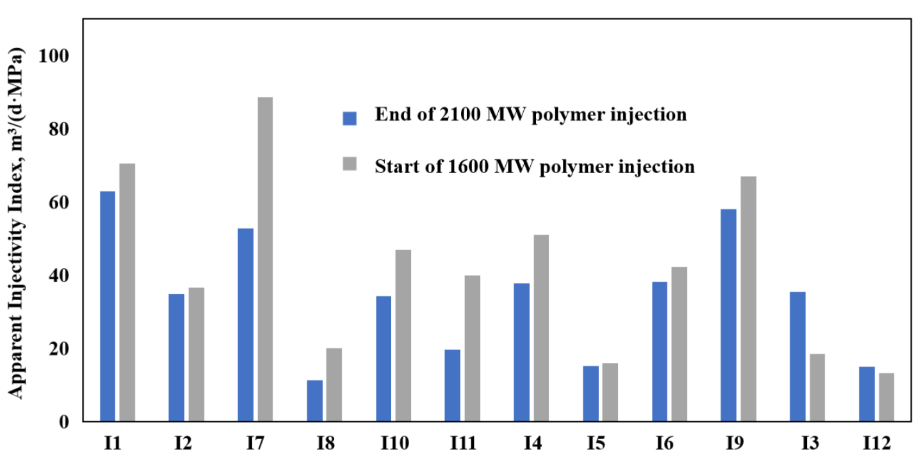

Figure 8—The apparent injectivity index before and after 16-million MW polymer injection for 12 injectors.

The sudden increased H_2S

At the beginning of 2024, a significant increase in Sulfate-Reducing Bacteria (SRB) was detected in the produced water system during platform monitoring, accompanied by the emergence of H_2S in production processes and observed fluctuations in polymer viscosity. The scheme required a wellhead viscosity of 30 mPa·s at polymer concentration of 1200 mg/L. The viscosity measurement results can be found in Fig. 9. During field testing in 2023, all three viscosity measurements of the polymer solution met the design specifications. However, in 2024, only one out of five random field tests achieved the target values due to the influence of H_2S and limitations in the sampling methodology.

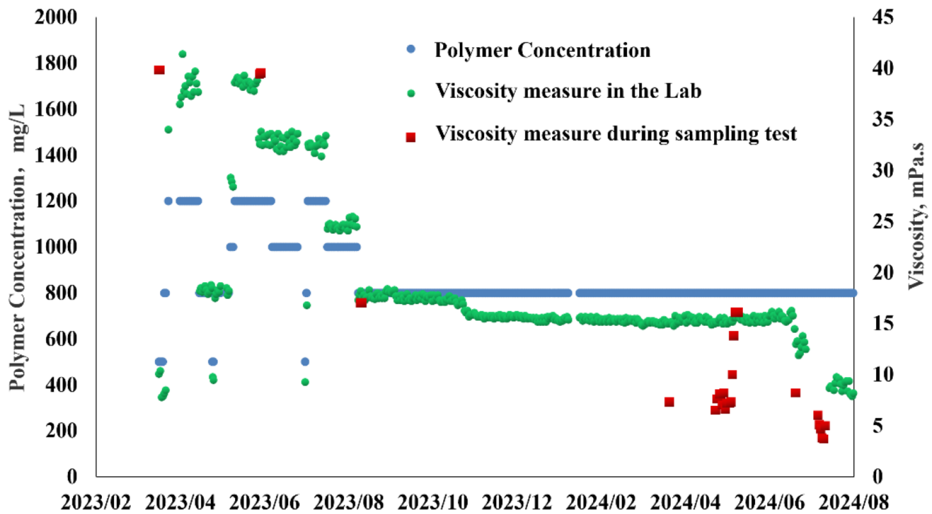

Figure 9—Viscosity measurement from the lab and platform sampling.

Numerical simulation inversion was applied to history-match the actual water cut of chemical flooding stage to determine the in-situ polymer viscosity. The history matching results is shown in Fig. 10. The water cut curve achieved optimal matching when the polymer viscosity was set to be 2.8 cp. This implies that due to the influence of H_2S, the actual in-situ polymer viscosity was 2.8 cp, which is significantly lower than the designed value of 8 cp to ensure effective oil displacement performance.

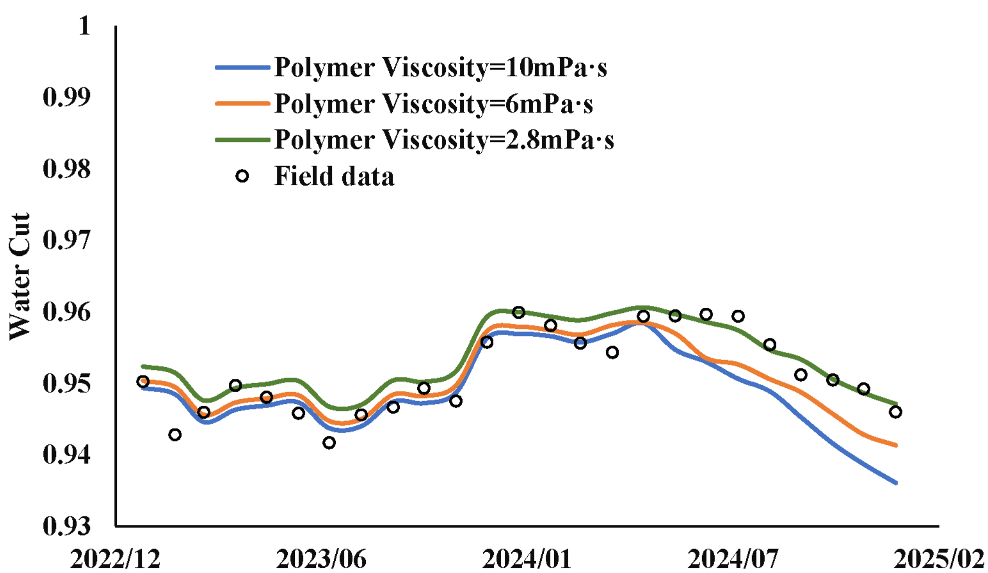

Figure 10—Numerical simulation result for water cuts with different polymer viscosity.

The formation of H_2S in oilfields is predominantly temperature-dependent. It typically results from thermochemical processes when the temperature is between 80 and 350°C, while founds from bacterial processes when the temperature is below 80°C. In this oilfield, the reservoir temperature ranges from 67 to 74°C provides optimal conditions for microbial proliferation. The degradation mechanism involves hydrolysis of H_2S into sulfides, where the resulting oxygen-sulfur system generates free radicals that induce the chain scission of polymer, consequently reducing the MW and viscosity of the polymer solution. Although the SRB are incapable of directly degrading high-molecular-weight polymers, the shorter-chain polymers formed after H_2S-induced viscosity reduction become more susceptible to microbial utilization. This process establishes a cyclical degradation pathway where SRB metabolize these polymer fragments to generate additional H_2S, thereby creating a self-perpetuating system of polymer degradation and viscosity reduction.

To address this problem, nitrogen displacement equipment was implemented into the polymer injection system to partially remove H_2S, while oxygen barrier measures were carried out to minimize the oxygen-induced viscosity reduction during polymer solution preparation. To further control oxygen exposure, thiourea was added to both the polymer solution and injection water to scavenge oxygen during the injection process, ensuring that the wellhead polymer viscosity met the requirement. However, due to the operational impacts on the water injection system caused by thiourea and potential risks associated with underground H_2S generation, only a minimal dosage of thiourea was incorporated into the polymer formulation. Through comprehensive treatment measures, the platform monitoring demonstrates a consistent downward trend in H_2S concentrations across both surface facilities and wellstreams. The H_2S concentration in the production system has been reduced from approximately 50 ppm to below 15 ppm, with all producing wells maintaining H_2S levels under 20 ppm. The monitored H_2S concentration in two producers is shown in Fig. 11. Moreover, the sulfide content has been effectively controlled at around 0.4 ppm. The solution of polymer with 21-million MW at 1200 mg/L concentration maintained stable wellhead viscosity above 22 mPa·s.

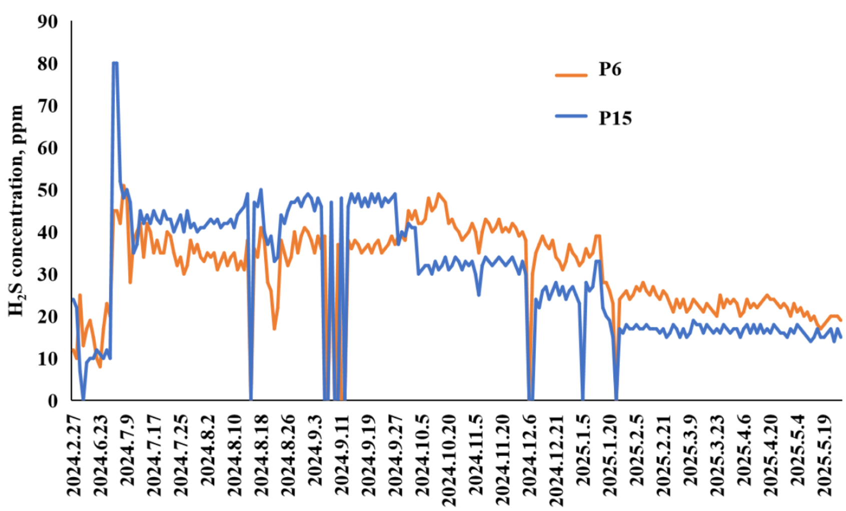

Figure 11—H_2S concentrations in two typical producers.

Performance of the Field Trial

The oil rate and water cut curves are shown in Fig. 12. As of mid-June 2025, the oilfield has maintained stable oil production at approximately 213 m^3/day, peaking at 232 m^3/day, with water cut reduced by about 1% compared to that of the stage before chemical flooding. The oil production rate has consistently exceeded the waterflooding baseline of 193 m^3/day for over 5.5 months.

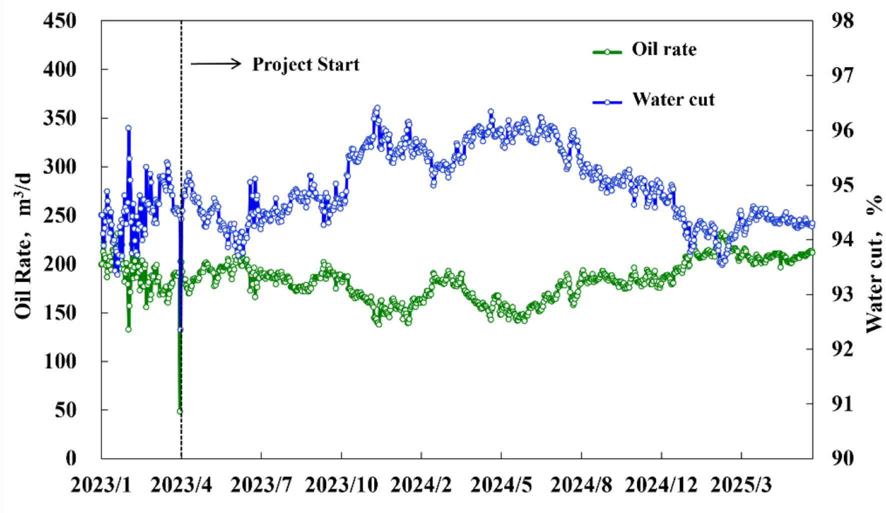

Figure 12—Oil rate and water cut of the field.

The polymer consumption is a key factor affecting the field performance during chemical flooding. The water cut versus polymer consumption is shown in Fig. 13. The delayed chemical flooding initiation, high-pressure induced insufficient injection, and polymer viscosity reduction have resulted in a great difference between actual performance and scheme prediction in the early stage of the project. However, after the treatments of well injectivity and H_2S, the recent performance has progressively approached the projected forecast.

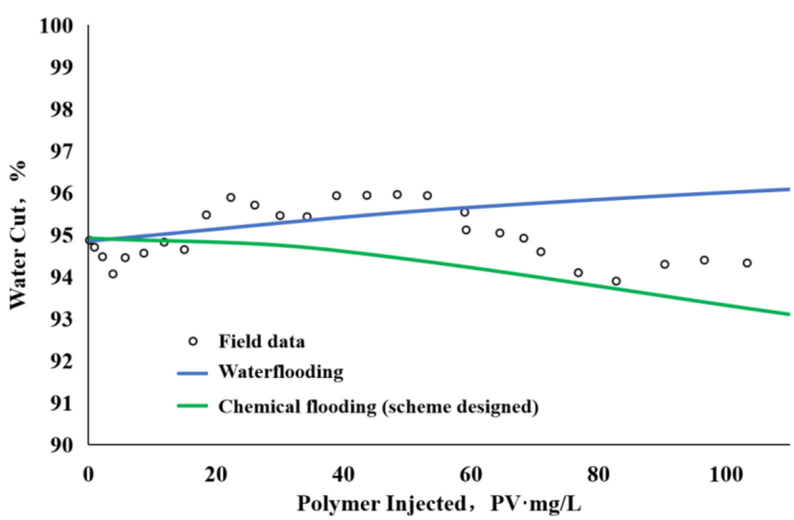

Figure 13—Water cut versus polymer consumption for the designed scheme and the actual field.

The evaluation criteria for production getting responded by the chemical flooding require sustained incremental oil production for more than 3 months by accounting for the decline rate of waterflooding, along with a maintained water cut reduction of at least 1% for more than 3 months. Based on this standard, 11 out of 17 production wells have shown decent oil incremental response. The responding wells demonstrate both decreased water cut and increased oil production compared to that of the stage right before chemical flooding. All wells in the central region performed greatly that affected by polymer injection. The non-responding wells are primarily horizontal wells located in the edge area of the oilfield. Despite a mature oilfield, those horizontal wells were only put into production two years prior to polymer injection. Therefore, different decline rates were applied for new and old wells when calculating incremental oil production.

Based on the initial production rates and decline trends before chemical flooding, the cumulative incremental oil was calculated to be 1.69 $\times 10^4$ m^3 using decline curve analysis (DCA). The actual and predicted waterflooding oil production curves for old wells and new wells are shown in Fig. 14.

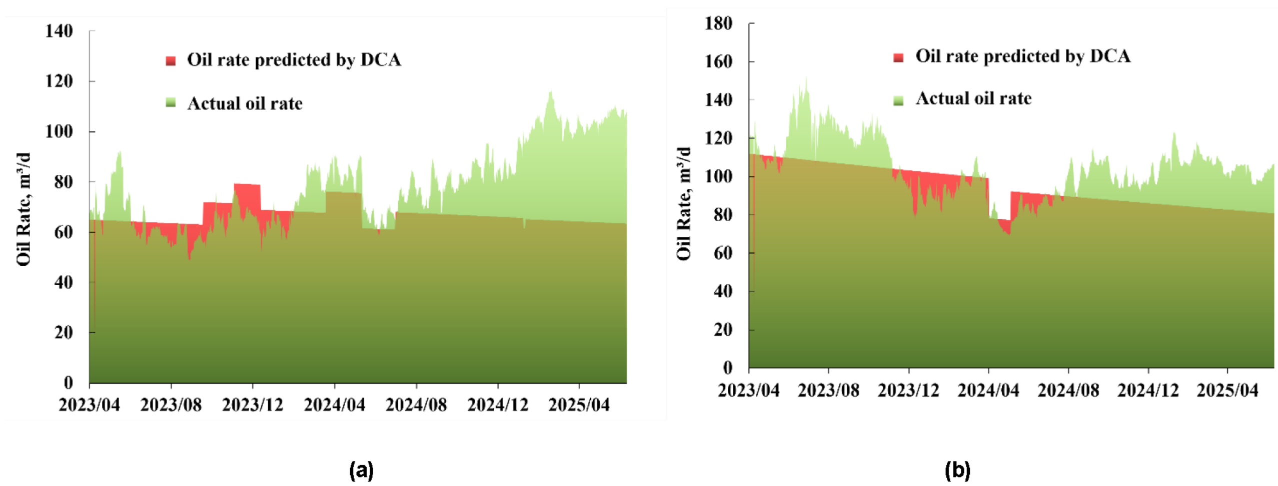

Figure 14—Actual oil rate and predicted waterflooding oil rate for (a) old wells; (b) new wells.

Conclusions

1. Acidizing and re-perforation treatments can effectively remove plugging from near-wellbore and wellbore region. Significant injectivity improvement, with increased water injectivity index was observed after well treatments. The alternating injection of polymers with 16-million MW and 21-million MW also achieved great performance on mitigating the injection pressure rise. Under non-plugging conditions in polymer injection wells, the lower MW polymer provides measurable injectivity improvements, effectively delaying the rate of pressure buildup during polymer injection.
2. The sudden increase in SRB concentration within the water system triggered H_2S generation, subsequently causing polymer degradation and viscosity reduction. Numerical simulation inversion revealed an in-situ polymer viscosity of 2.8 cP. The measures of nitrogen blanketing for oxygen isolation and thiourea addition to the polymer formulation effectively controlled H_2S concentrations in both production system and wellbores, restoring wellhead polymer viscosity to normal specifications.
3. Despite challenges including high-pressure induced insufficient injection and H_2S induced polymer degradation, the implement of DCF with different slugs demonstrated a decent oil production performance, with an incremental oil production of 1.69 $\times 10^4$ m^3 calculated using DCA.

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